Multimedia Signal processing

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1. **Question (8 pts)**:
   Consider a (somewhat simplified) document collection, which is described by the following term-document matrix

$$A = \begin{bmatrix} 10 & 8 & 2 & 8 & 0 & 2 \\ 0 & 1 & 4 & 3 & 2 & 0 \\ 1 & 3 & 0 & 1 & 4 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}, \tag{1}$$

   where $A_{i,j}$ represents the number of times term i (row index) appears in document j (column index).
   (a) Compute the number of bits needed to Huffman code each document (assigning one codeword to each term).
   (b) Compute the number of bits needed to arithmetic code each document.
2. **Question (5 pts)**
   (a) Illustrate the principles of transform coding.
   (b) Describe how the bitrate is allocated to each transform coefficient.
3. **Question (5 pts)**
   (a) Illustrate the block diagram of an analysis-by-synthesis speech coder.
   (b) Illustrate how it is possible to take into account human perception in such coder.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**

   (a) Consider a memoryless source X, whose p.d.f. is denoted by $p_X(x)$, with variance σ_x^2

   - [T] [F] The distortion introduced by lossy coding decreases linearly with rate.
   - [T] [F] $D_U(R) \geq D_{LM}(R)$, where U and LM denote, respectively, uniform and Lloyd-Max quantization.
   - [T] [F] If $p_X(x)$ is uniform, then $D_U(R) = D_{LM}(R)$
   - [T] [F] The rate-distortion function depends only on σ_x^2, and it does not depend on the shape of the p.d.f. $p_X(x)$

   (b) Consider an image coding system.

   - [T] [F] Color components are separately encoded in the RGB colorspace
   - [T] [F] The coding efficiency increases by increasing the block size.
   - [T] [F] When using threshold quantization it is possible to determine a-priori the number of bits needed to encode an image.

   (c) Let I_c and I_r denote, respectively, the current and the reference frame used in motion estimation and compensation

   - [T] [F] $PSNR(DFD_1(I_c, I_r)) \geq PSNR(DFD_2(I_c, I_r))$, where DFD_1 and DFD_2 denote, respectively, the displaced frame difference computed using full-pixel and half-pixel precision, respectively.
   - [T] [F] $PSNR(DFD(I_c, I_r; B_1)) \geq PSNR(DFD(I_c, I_r; B_2))$, if $B_1 > B2$
   - [T] [F] The computational complexity of motion estimation is higher than that of motion compensation.

5. **Question (Matlab - 10pts)**

   You are asked to implement a PCM coder based on a compander-expander.

   - Read the input signal *speech.wav* and show the histogram of the source.
   - Estimate the variance of the source σ_x^2 and fit a Laplacian distribution

   $$p(x) = \frac{\lambda}{2} e^{-\lambda|x|} \tag{2}$$

   where $\lambda = \sqrt{2}/\sigma_x$

   - Quantize the signal with a compander-expander based on the following function.

   $$y = T[x] = \begin{cases} \frac{1-e^{-\lambda x}}{2} & x \geq 0 \\ \frac{e^{-\lambda x}}{2} & x < 0 \end{cases} \tag{3}$$

   - Plot the rate-distortion function (SNR vs. rate) by varying the quantization step size Δ (Hint: estimate the rate based on the entropy of the quantization symbols)